# AMPERE DEVELOPMENT

Power Testing & Load Bank Solutions

# Load Bank Selection & Sizing Guide

How to Choose kW, Voltage, Power Factor, Type, Form Factor & Cables

Wrong capacity, wrong voltage, or the wrong load type wastes time and money — and can invalidate a test. This guide walks through the decisions that matter so you can specify (or rent) the right load bank the first time. **Use it as a checklist before you call for a quote.**

## What You Will Decide

1. **Capacity (kW / kVA)** — how much load do you need?
2. **Voltage & frequency** — match the unit under test
3. **Load type** — resistive, reactive, capacitive, or combined
4. **AC vs DC** — and when electronic / programmable loads matter
5. **Form factor** — indoor, outdoor, portable, trailer, container
6. **Cables & connections** — length, ampacity, connectors
7. **Heat rejection & site constraints** — where the heat goes

## 1. Capacity: How Much kW (and kVA) Do You Need?

Start from the nameplate of the equipment under test — not from a guess. For generators, the critical figure is usually **nameplate kW** (real power). For alternators and many acceptance tests you also care about **kVA** (apparent power).

- **Full-capacity / acceptance test:** Plan for 100% of nameplate kW (and often 100% kVA if reactive testing is required).
- **NFPA 110-style annual tests:** Follow the required percentages (e.g. 50% then 75% of nameplate kW). You still need a bank that can reach the highest step.
- **Wet-stacking cleanup:** Often needs sustained higher load — size for the cleanup target, not only the monthly minimum.
- **Data-center IST:** Size to the design IT load (or the portion under test), plus margin if cooling is tested in parallel.
- **Derating:** High altitude and high ambient temperature can reduce available engine power — confirm site conditions.

**Tip:** If you only have kVA and power factor, kW ≈ kVA × PF. Example: 1000 kVA at 0.8 PF ≈ 800 kW of real power.

## 2. Voltage & Frequency

The load bank must be compatible with the voltage and frequency of the source under test. Mismatches are unsafe and can damage equipment.

- Common low-voltage AC: 208 V, 240 V, 480 V, 600 V (3-phase); single-phase 120/240 V for smaller sets
- Medium-voltage: 4160 V, 12.47 kV, 13.8 kV and others — requires MV-rated banks and specialized cables/safety
- Frequency: 60 Hz (North America typical) or 50 Hz; some aviation applications use 400 Hz
- Multi-voltage banks exist, but confirm the rating at **your** operating voltage — capacity can change with voltage selection

**Always state the exact voltage(s) and whether the source is single-phase or three-phase when requesting a quote.**

## 3. Power Factor & Load Type: Resistive, Reactive, Capacitive

Not all “load” is the same. The type of load bank determines whether you are stressing only real power (kW) or also reactive power (kVAR) and power factor.

| Type | What It Simulates | Typical Use |
| --- | --- | --- |
| Resistive (PF ≈ 1.0) | Pure real power (kW). Heat only. | Generator kW capacity, wet stacking, many UPS tests, basic commissioning |
| Resistive-Reactive (e.g. 0.8 PF lag) | Real + lagging reactive (kW + kVAR) | Full kVA / alternator tests, more realistic facility loads |
| Capacitive (leading PF) | Leading reactive load | Certain UPS, inverter, and power-factor correction scenarios |
| Combined / hybrid | Selectable or stepped PF | Flexible testing when both pure kW and PF tests are needed |

### When Resistive Is Enough

- Monthly or annual diesel exercise focused on engine load and exhaust temperature
- Simple generator kW capacity checks
- Many battery and resistive UPS discharge tests

### When You Need Reactive or Combined

- Acceptance testing of alternators to nameplate kVA at specified power factor (often 0.8 lagging)
- Simulating real building or industrial loads that are not purely resistive
- Data-center and facility tests where power-factor behavior matters

If the specification or commissioning plan calls for a power-factor target, say so early. A resistive-only bank cannot produce lagging or leading VARs.

## 4. AC vs DC (and Electronic Loads)

- **AC load banks** — standard for generators, most UPS output testing, facility AC distribution.
- **DC load banks** — battery strings, telecom –48 V, UPS DC buses, solar/BESS DC sides, EV-related DC, higher-voltage data-center DC (e.g. 380/400 V, 800 V class).
- **Electronic / programmable loads** — precise CC/CV/CP/CR modes, dynamic steps, and profiles for power-supply, inverter, and R&D; testing.

**Do not assume an AC bank can test a DC system (or vice versa). Voltage class and polarity matter as much as capacity.**

## 5. Form Factor: Indoor, Outdoor, Portable, Trailer, Container

Physical package drives site logistics, weather protection, and how much capacity you can deploy in one footprint.

| Form | Best For | Watch-outs |
| --- | --- | --- |
| Portable / wheeled | Small–mid kW, indoor or short outdoor moves | Limited capacity; cable management |
| Skid-mounted | Mid capacity, forklift moves, temporary outdoor | Need forklift / crane access |
| Trailer-mounted | Field generator tests, multi-site, fast mobilize | Road access, hitch, weight limits |
| Container / ISO | Multi-MW, longer projects, weather protection | Crane / trailer delivery; footprint |
| Indoor rack / server-sim | Data-center hall, cooling + power IST | Floor loading, hot-aisle planning |
| Permanent / station | Very high utilization, fixed voltage | Capital, storage, maintenance |

Outdoor units need clear intake and exhaust paths. Indoor units need HVAC capacity to handle the heat they reject.

## 6. Cables, Connectors & Length

Cables are part of the system. Undersized or too-short cables create voltage drop, heat, and safety issues.

- **Ampacity:** Must support full test current at your voltage ($I \approx kW \times 1000 / (\sqrt{3} \times V \times PF)$ for 3-phase).
- **Length:** Longer runs need larger conductors or accept more drop. State the required length early.
- **Connectors:** Cam-lok, bus bar, sequential interlocking, MV terminations — match the site interface.
- **Quantity:** Parallel cables are common at high current; plan for the full set, not a single lead.
- **Rental packages:** Many rentals include a cable set — confirm length and connector type in the quote.

Always tell the supplier the distance from load bank to connection point and the connector standard on the unit under test.

## 7. Heat Rejection & Site Constraints

A load bank turns electrical power into heat. That heat has to go somewhere without damaging people, buildings, or other equipment.

- **Air-cooled banks:** Require clear intake and high-temperature exhaust path. Do not point exhaust at people, soft materials, or air intakes of other equipment.
- **Exhaust temperatures:** Can exceed 200 °F (93 °C) at the outlet — treat as a burn and fire hazard zone.
- **Indoor testing:** Building HVAC or temporary exhaust must handle the full thermal load. Server-sim / rack banks are designed for data-hall airflow patterns.
- **Liquid-cooled options:** Used when heat must be absorbed into a cooling loop (e.g. some high-density data-center tests) rather than dumped to ambient air.
- **Noise:** Fans are loud; plan hearing protection and neighbor/site rules.
- **Footprint & weight:** Confirm pad, trailer, or floor loading capacity before delivery.

**Site checklist: Clearance for airflow, exhaust direction, floor/pad capacity, crane or forklift access, weather protection, and exclusion zone for personnel.**

## 8. Quick Selection Path

1. **What are you testing?** Generator / UPS / battery / data hall / MV gear / renewable inverter / other.
2. **What capacity and voltage?** Nameplate kW (and kVA if needed), exact voltage, 50/60/400 Hz, 1Ø or 3Ø.
3. **What power factor?** Resistive only, or 0.8 lag (or other) reactive required?
4. **AC or DC?** Any programmable/electronic requirement?
5. **Where will it sit?** Indoor / outdoor / trailer / container / rack — access and heat path.
6. **How far is the connection?** Cable length, connector type, ampacity.
7. **How long and how often?** Days vs weeks; one site vs multi-site — affects rental vs own and form factor.

## 9. What to Send for an Accurate Quote

- ■ Application (generator acceptance, NFPA test, IST, battery discharge, etc.)
- ■ Required kW / kVA and target power factor
- ■ Voltage(s), frequency, phase
- ■ AC or DC (and voltage class if DC)
- ■ Preferred form factor (portable, trailer, container, rack, etc.)
- ■ Cable length and connector preference
- ■ Site location, dates, indoor/outdoor, access notes
- ■ Heat-rejection constraints or liquid-cool preference
- ■ On-site tech support needed? Yes / No

**Ready to size the right bank?**

Send the checklist above — or call with nameplate data and test goal. We will match capacity, type, form factor, and cables to your job.

**888-345-LOAD · ampere.dev · sales@ampere.dev**

*This guide is for planning and specification. Always confirm final selection against manufacturer ratings, site conditions, and applicable codes. Ampere Development can help translate your nameplate and test plan into a concrete rental package.*